

EXPLORATORY DATA ANALYSIS, STATISTICAL ANALYSIS AND MACHINE LEARNING USING THE TITANIC DATASET

1. Introduction

Data science combines statistical methods, programming, and machine learning techniques to extract meaningful information from data and support decision-making. Exploratory Data Analysis (EDA) is an essential stage of the data science process because it helps analysts understand the structure of a dataset, identify patterns, detect anomalies, and prepare data for predictive modelling.

The Titanic dataset is one of the most widely used datasets for learning data science and machine learning. It contains demographic and travel information about passengers aboard the RMS Titanic, including variables such as age, gender, passenger class, fare, family relationships, and survival status. Using this dataset, it is possible to investigate the factors that influenced passenger survival and develop predictive models.

The main objective of this project is to perform exploratory data analysis, statistical analysis, and machine learning using Python. The project applies various Python libraries, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-Learn, to clean, visualize, analyze, and model the Titanic dataset. The study also demonstrates the practical application of Logistic Regression in predicting passenger survival.

2. Dataset Description

The dataset used in this project is the Titanic dataset obtained from Kaggle. It contains information about passengers who travelled aboard the Titanic during its maiden voyage.

The dataset consists of 891 observations and 12 variables. The variables include PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked.

The target variable for the machine learning model is **Survived**, where:

- 0 = Passenger did not survive
- 1 = Passenger survived

The remaining variables provide demographic and travel-related information that can be used to predict survival.

3. Methodology

The project was carried out in several stages.

Data Acquisition

The Titanic dataset was downloaded from Kaggle and imported into Jupyter Notebook using the Pandas library.

Data Cleaning

Data preprocessing involved identifying missing values and duplicate records.

The Age column contained 177 missing values. These missing values were replaced using the median age because the median is less sensitive to outliers.

The Embarked column contained two missing values. These values were replaced using the mode (most frequent embarkation port).

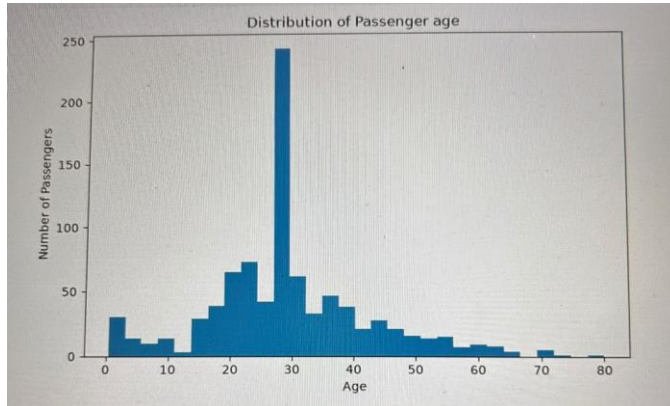
The Cabin column contained 687 missing values, representing more than three-quarters of the dataset. Therefore, the column was removed because it contained insufficient information for meaningful analysis.

Duplicate records were checked using the duplicated() function. No duplicate records were found.

Data Visualization

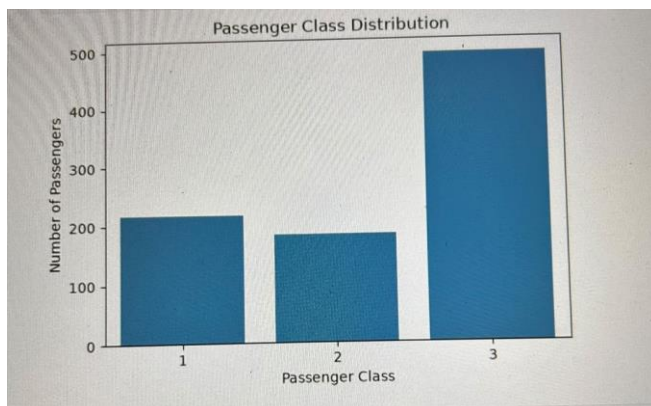
Several visualizations were created to understand the characteristics of the dataset, including:

- **Histogram of passenger ages**



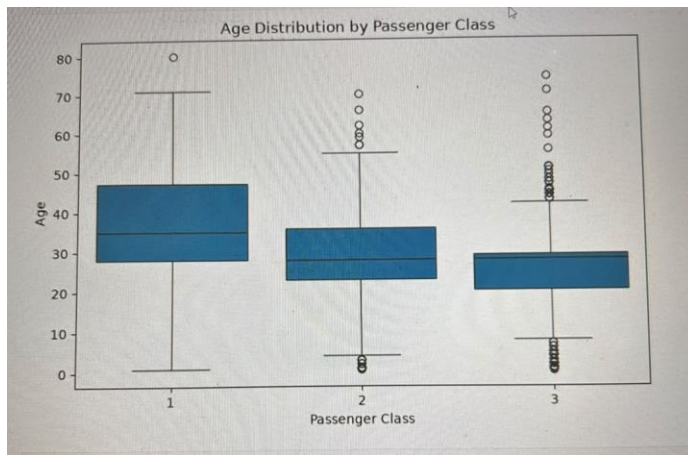
This histogram shows that most passengers were between 20 and 40 years old. There were relatively fewer children and elderly passengers.

- **Passenger class distribution**



Most passengers travelled in Third Class, followed by First Class, while Second Class had the fewest passengers.

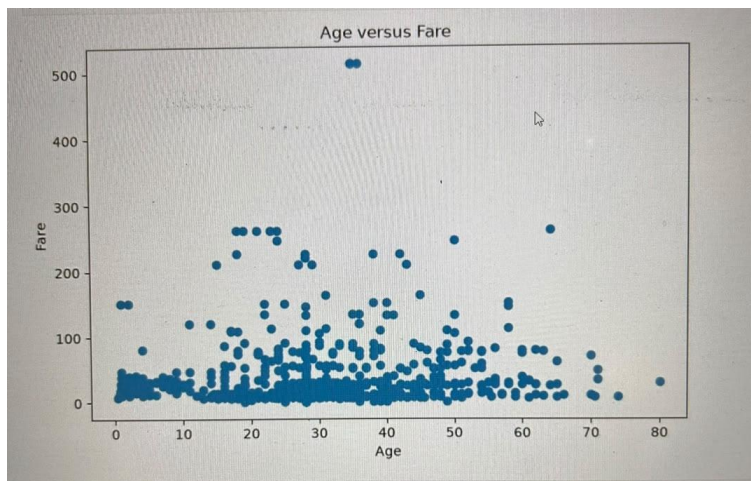
- **Boxplot of age by passenger class**



This boxplot shows the distribution of passenger ages within each class.

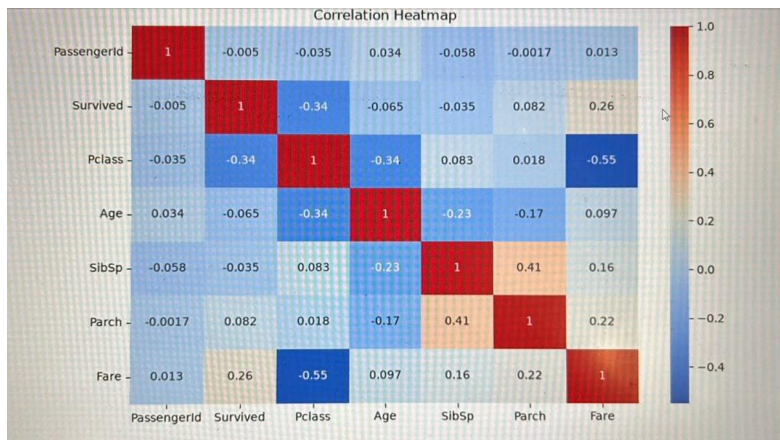
It also highlights outliers and differences in age spread across the passenger classes.

- **Scatter plot of Age versus Fare**



Most passengers paid relatively low fares regardless of age. A few passengers paid significantly higher fares, appearing as outliers.

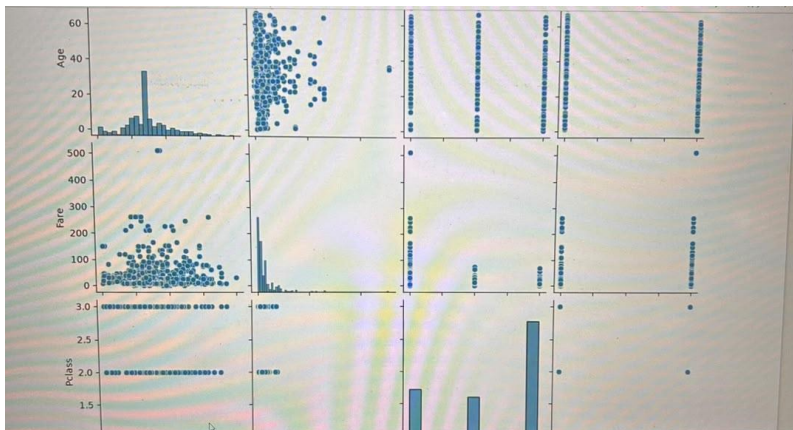
- Correlation heatmap



The heatmap shows the strength and direction of relationships between numerical variables.

Darker colours indicate stronger positive or negative correlations.

- Pairplot of selected numerical variables



This pairplot illustrates relationships among age, fare, passenger class, and survival.

It helps identify trends, clusters, and potential correlations between variables.

Statistical Analysis

Descriptive statistics, frequency distributions, and correlation analysis were performed to summarize the dataset and identify relationships among variables.

Machine Learning

A Logistic Regression classifier was developed using Scikit-Learn.

The selected predictor variables were:

- Passenger Class
- Sex
- Age
- Fare
- Number of Siblings/Spouses
- Number of Parents/Children
- Embarked

The dataset was divided into training data (80%) and testing data (20%). The model was trained using the training dataset and evaluated using the testing dataset.

4. Results

The exploratory data analysis showed that most passengers travelled in Third Class. The average passenger age was approximately 29 years, while the average ticket fare was 32.20.

The correlation analysis revealed that Passenger Class had a moderate negative relationship with Fare (-0.5495), indicating that passengers in higher-numbered classes generally paid lower fares. A moderate positive relationship (0.4148) existed between SibSp and Parch, suggesting that many passengers travelled with family members.

The Logistic Regression model achieved an accuracy of **81.01%** on the testing dataset.

The confusion matrix showed that:

- 90 non-survivors were correctly classified.
- 55 survivors were correctly classified.

- 15 passengers were incorrectly predicted to have survived.
- 19 passengers were incorrectly predicted not to have survived.

The classification report indicated balanced model performance, with an overall weighted F1-score of approximately 0.81.

5. Discussion

The exploratory analysis demonstrated that passenger characteristics such as ticket class, fare, and gender were associated with survival. Passengers travelling in First Class generally paid higher fares and had better survival outcomes than those travelling in Third Class.

The statistical analysis further confirmed relationships among the variables. Passenger class and fare exhibited the strongest negative correlation, while family-related variables (SibSp and Parch) showed the strongest positive correlation.

The Logistic Regression model performed well, correctly classifying approximately 81% of the passengers. Although the model produced good results, some passengers were misclassified, indicating that additional variables or more advanced machine learning algorithms could improve prediction accuracy.

6. Conclusion

This project successfully demonstrated the application of exploratory data analysis, statistical analysis, and machine learning using the Titanic dataset.

The data cleaning process improved data quality by handling missing values appropriately. Visualizations provided valuable insights into passenger characteristics, while statistical analysis revealed important relationships among the variables.

The Logistic Regression model achieved an accuracy of 81.01%, demonstrating that machine learning techniques can effectively predict passenger survival based on demographic and travel information.

Overall, the objectives of the project were achieved, and the study provided practical experience in applying Python for real-world data science tasks.

7. Recommendations

- Compare Logistic Regression with other machine learning algorithms such as Random Forest and Decision Trees.
- Perform feature engineering to create additional predictor variables.
- Apply cross-validation to improve model evaluation.
- Investigate additional datasets for comparison and further study.

8. References

- Kaggle. Titanic Machine Learning Competition Dataset.
- McKinney, W. *Python for Data Analysis*. O'Reilly Media.
- Pedregosa, F., et al. *Scikit-Learn: Machine Learning in Python*.
- The Pandas Documentation.
- The Seaborn Documentation.
- The Matplotlib Documentation.